BME 5910 - HW 1

Problems 1.2 and 1.4

Author: Garrett Kunkler

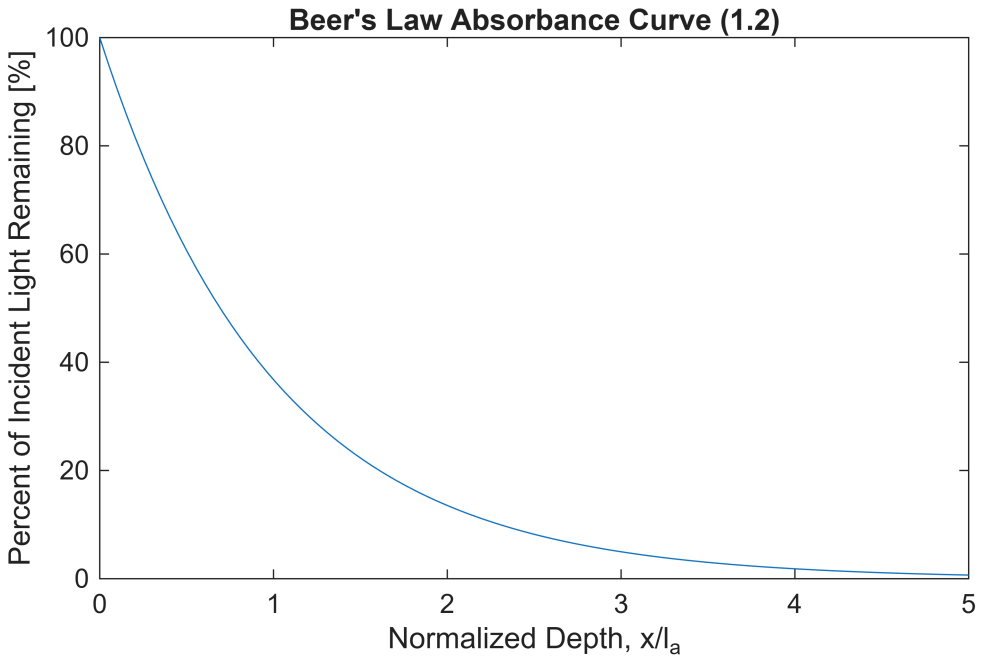
Aids:

- Textbook: Biomedical Optics: Principles and Imaging by Lihong Wang and Hsin-I Wu

```
% Knowns
mu_a = 1;
mu_s = 2;
L = 5;
I_0 = 1;
x = linspace(0,L,100);
```

```
% Calculations
I_a = I_0 * exp(-mu_a*x);
I_s = I_0 * exp(-mu_s*x);
percent_remaining_absorbed = I_a/I_0 * 100;
percent_remaining_scattered = I_s/I_0 * 100;
```

```
% Plot absorbance
plot(x*mu_a, percent_remaining_absorbed)
xlim([0,5])
xlabel("Normalized Depth, x/l_a")
ylabel("Percent of Incident Light Remaining [%]")
title("Beer's Law Absorbance Curve (1.2)")
```



```
% Plot scattering
plot(x*mu_s, percent_remaining_scattered)
xlim([0,5])
xlabel("Normalized Depth, x/l_s")
ylabel("Percent of Incident Light Remaining [%]")
title("Beer's Law Scattering Curve (1.4)")
```

